# Christian Sassi

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## Education

**Master's Degree in Computer Science, Cybersecurity - University of Trento** — Trento, Italy
*Graduated with a final grade of 110/110 cum laude.* — *Sep. 2023 – Dec. 2025*

**Bachelor's Degree in Computer Engineering - University of Trento** — Trento, Italy
*Graduated with a final grade of 108/110.* — *Sep. 2020 – Sep. 2023*

## Experience

**Bug Bounty Program Operator** — Feb. 2026 – Present
*Huawei* — *Munich, Germany*

- Conduct **mobile security** research on private **bug bounty programs**.

**Software Developer** — Nov. 2025 – Jan. 2026
*Energee3* — *Reggio Emilia, Italy*

- Designed a secure **Python** backend that orchestrates authenticated access to multiple external databases through an **encrypted local metadata layer**.
- Built a web-based **SQL client** (**HTML**, **CSS**, **JavaScript**) for safe query execution with efficient **connection management** and no external dependencies.

**Researcher Intern - Cybersecurity** — Mar. 2025 – Dec. 2025
*Fondazione Bruno Kessler (FBK)* — *Trento, Italy*

- Engineered a **federated learning** pipeline by adapting the **DAICS** deep-learning anomaly-detection framework for **Industrial Control Systems (ICS)** security.
- Integrated **adaptive aggregation** and **client selection** from the **FLAD** framework to improve training efficiency.
- Benchmarked on the **SWaT** dataset, with preliminary results approaching the centralized DAICS baseline while preserving **data privacy**.

**Software Engineer Intern - Embedded Systems** — Feb. 2023 – Jun. 2024
*University of Trento* — *Trento, Italy*

- Validated **SPARK SR1020 Ultra-Wideband (UWB) IoT** devices through **specification-based testing**.
- Optimized **C** firmware for real-time **UWB** communication, reducing response time by **50%**.
- Developed **Python** tools to support firmware debugging and performance tuning.

## Projects

**MITRE Embedded Capture The Flag (eCTF) Competition** | *C*

- Competed in MITRE's **eCTF** national **firmware security** competition, where the team placed **13th of 116**.
- Wrote defensive **C** code in the embedded firmware: **memory zeroing**, **overflow checks**, and overflow-safe primitives.

**Security Testing Analysis for an E-commerce Platform** | *Java*

- Ran **static (SAST)** and **dynamic (DAST)** analysis of a **Java** platform with **SpotBugs** and **OWASP ZAP**.
- Identified **15+ vulnerabilities** including **SQLi**, **XSS**, and **CSRF**, triaging findings for remediation.
- Designed and applied **15+ targeted fixes** for the discovered issues.

**T.R.U.S.T. - Secure Chat Application** | *Rust*

- Built an **end-to-end encrypted** chat application in **Rust** (TUI), implementing **Signal** protocol primitives.
- Implemented **X3DH** key establishment, the **Double Ratchet** (forward secrecy), and authenticated **AES-GCM** encryption.

## Publications

**Real-time Musical Haptics with Ultra-wideband: A Study on Latency, Reliability, and Perception**
*L. Turchet, **C. Sassi**, D. Vecchia, G. Picco* - Published in IEEE Transactions on Haptics on January 6, 2025.

## Skills

**Spoken Languages**: Italian, English

**Programming Languages**: Python, C, C++, Java, Rust, JavaScript, HTML, CSS, Bash, Solidity, SQL

**Security**: Vulnerability Assessment, Web Security, Mobile Security, Network Security, Reverse Engineering, Binary Exploitation (Pwn), Injection Attacks, Cryptography, Federated Learning

**Security Testing**: SAST, DAST, OWASP Top 10, CVE, CWE